

5(a)	<i>any three from:</i> allows DNA replication to, occur many times / AW ; allows (some) cells to carry out, many / continuous / repeated, mitoses / cell cycles / cell divisions ; I cell replication prevents the loss of, genes / genetic information (from ends of chromosomes) ; ora A prevents loss of coding sequences I prevents loss of, DNA / genetic material AVP ; e.g. prevents fusion of chromosome ends prevents ends of chromosomes being recognised as damaged	3
5(b)(i)	U U A G G G ;	1
5(b)(ii)	leucine ; glycine ; ecf on asparagine and proline	2
5(b)(iii)	<i>any one from:</i> increase in number of different proteins produced ; A increase in number of different enzymes produced A increase in the number of different mRNA molecules in the cell <i>ref. to</i> ability of cell to, become differentiated / become specialised / take on particular function ; increase in growth of cell (in preparation for mitosis) ; production of cell organelles (in preparation for mitosis) ; AVP :	1